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W5-01: Persistent Storage Primer

COMPSCI4064 (H) and COMPSCI5088 (M)

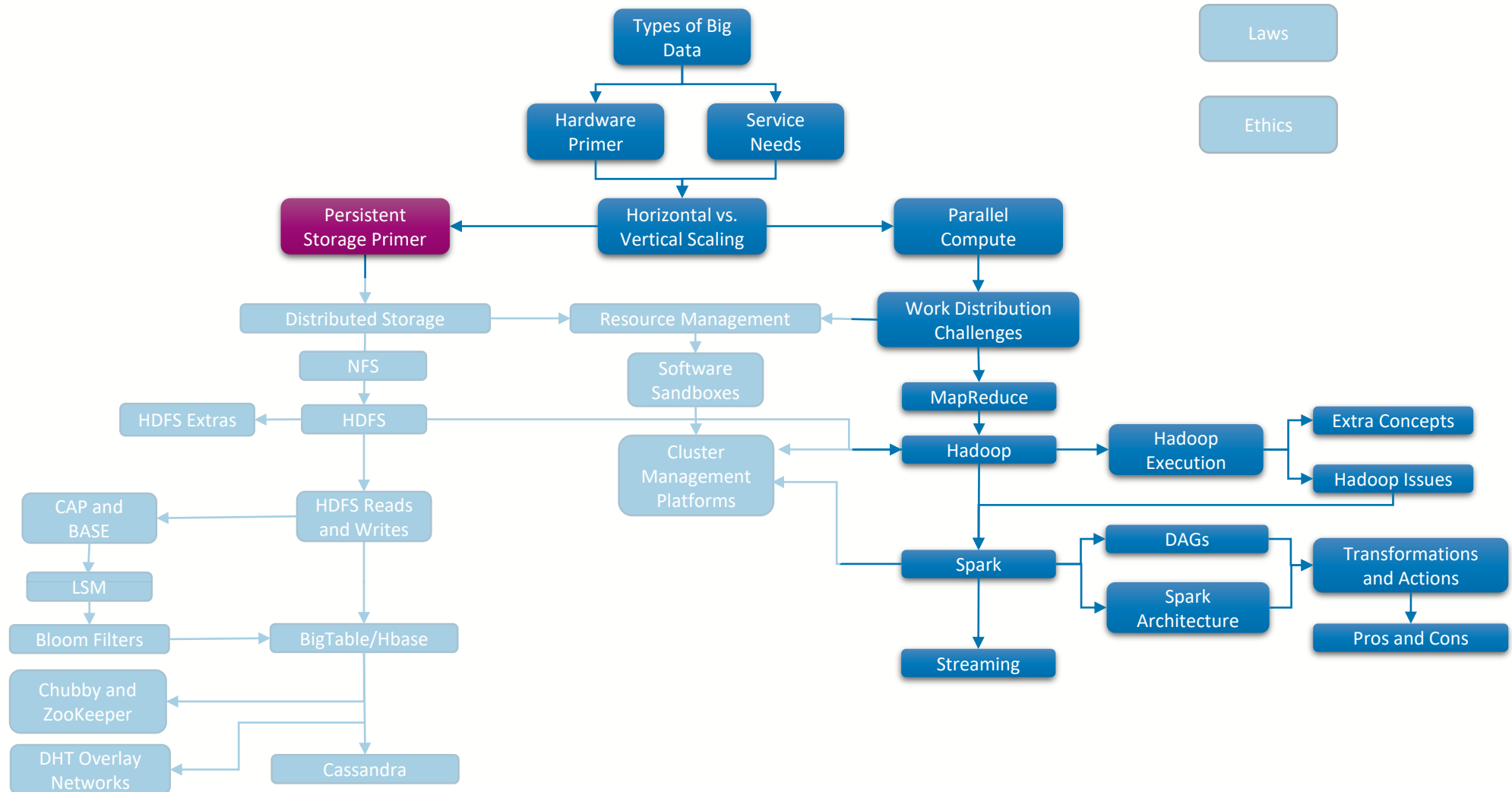
Dr. Richard McCreddie

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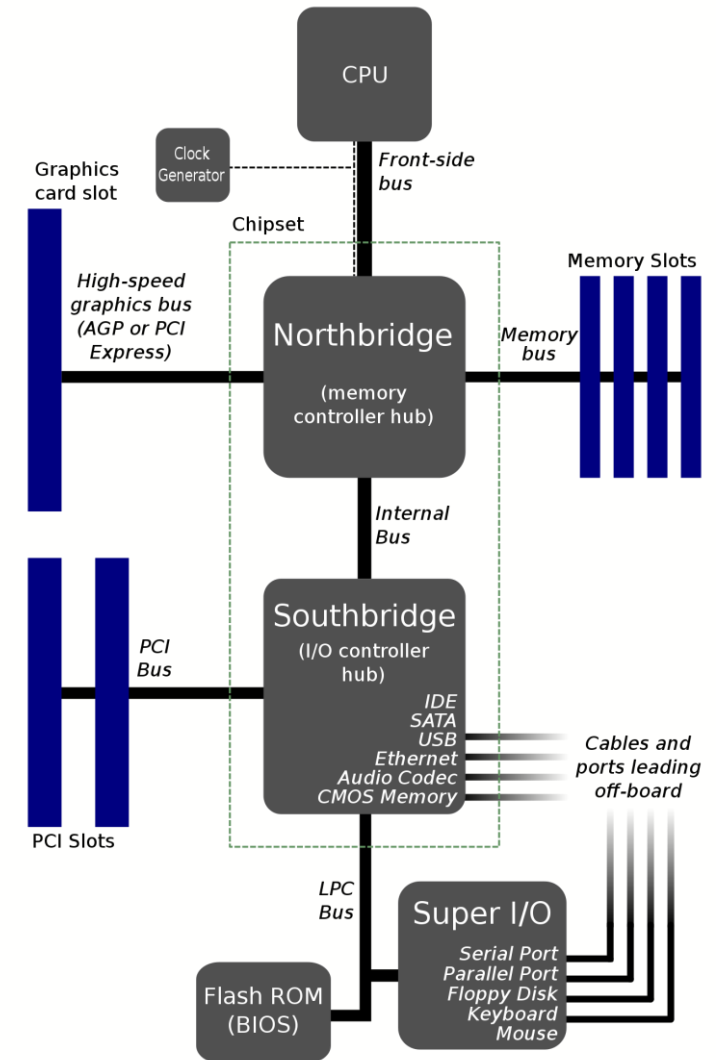


Where Are We?



A Quick Aside on Motherboard Connections

- Within a traditional computer, the various storage media are connected to the CPU via connections referred to as a 'bus'
 - The closer we get to the CPU the faster the connection will be
- Main Memory and PCIe are the closest connections to the CPU
- Ethernet (network), SATA and USB are more distant





Host Interfaces

- The storage connection is known as the host interface
- Wide range of interfaces available
 - Serial attached SCSI (SAS-3, 12.0 Gbit/s) – generally found on servers
 - Serial ATA and mSATA variant (SATA 3.0, 6.0 Gbit/s)
 - PCI Express 3 (PCIe 3.0 ×4, 31.5 Gbit/s)
 - PCI Express 4 (PCIe 4.0 ×4, 63 Gbit/s)
 - Fibre Channel (128 Gbit/s) – almost exclusively found on servers
 - USB (10 Gbit/s)
- Even if the underlying storage medium is faster, it can be bottlenecked by the host interface
 - This is why M.2 SSDs are usually connected to the PCI Express bus.

Hard Disk Drive

- Idea: Store data on **rigid platters** coated with **magnetic material**
 - Rapidly rotate the platter to get to the right data (e.g. 7,200 RPM)
 - Lasts for ~4-6 years
- Characteristics
 - Slow for Random Reads
 - Medium Speed for Block reads
 - Slow for Writes
 - Cheap
- Uses a small amount of DRAM as a volatile cache, backed with a battery to reduce latency
- Connects via SATA

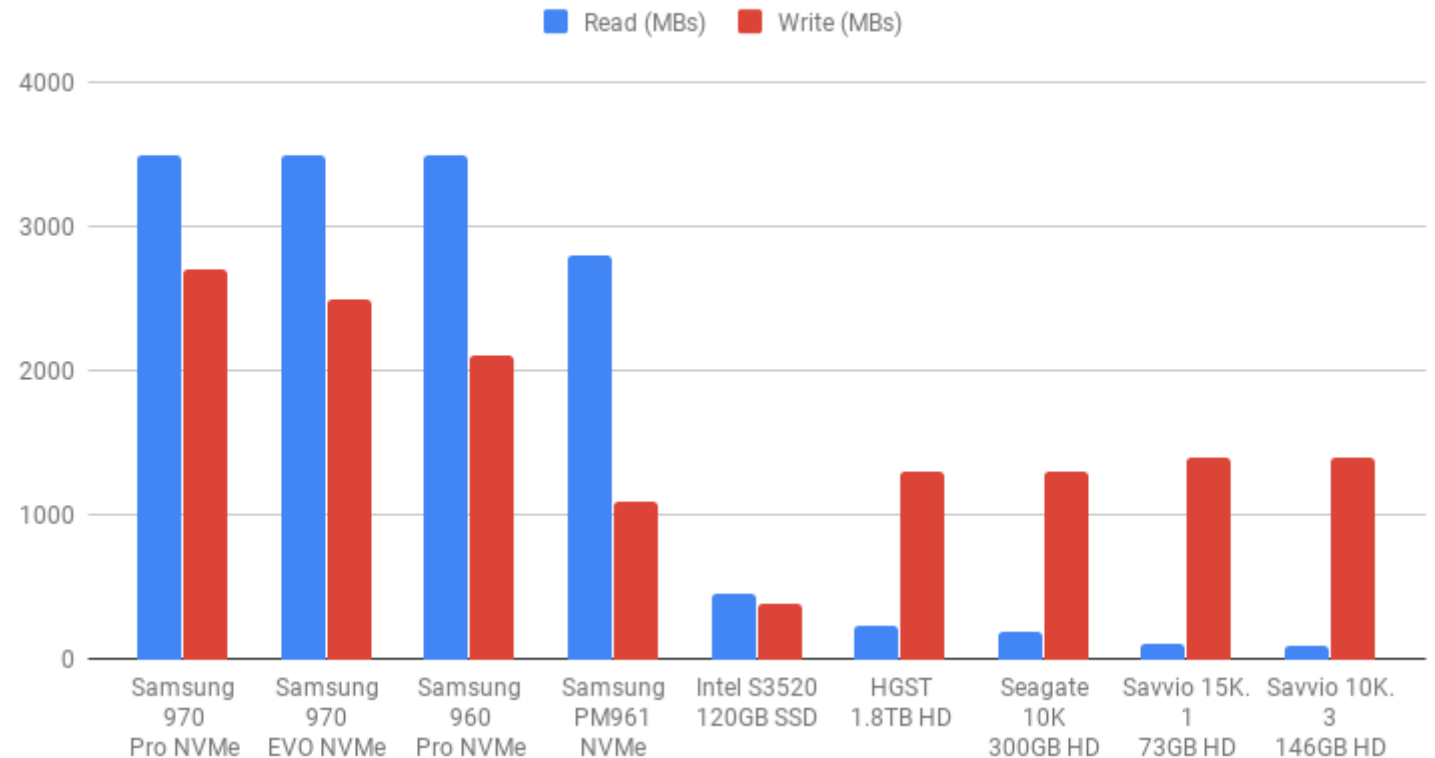




Solid State Drives

- Idea: Store data in **semiconductor cells**
 - Need to receive power periodically (once per 1-2 years) or experience data loss
- Not a new idea, conceived in 1978
 - Reached military use around 1995
 - Wide spread consumer use in the 2010's
- Have much faster **random read speeds** than HDDs
 - Also uses a small amount of DRAM as a volatile cache, backed with a battery
- Connects via SATA or PCIe

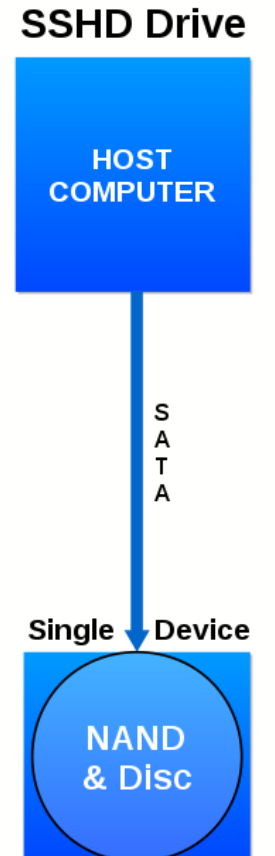
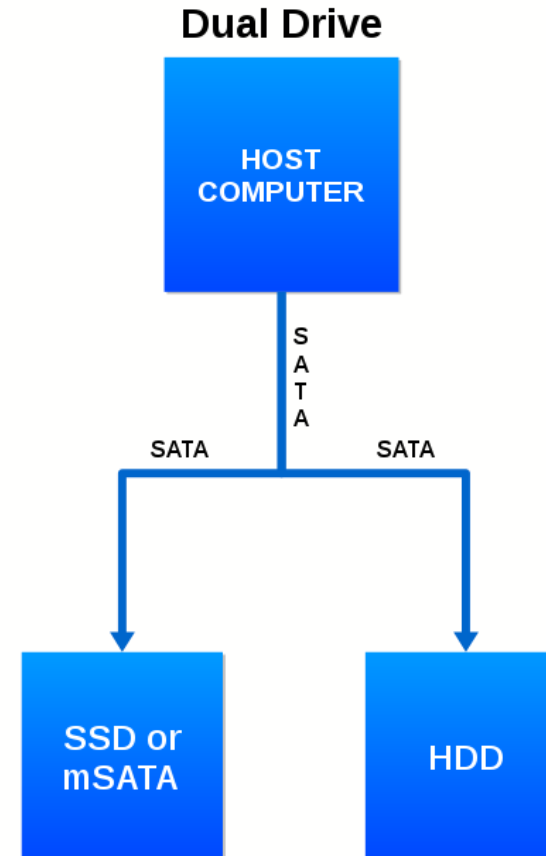
Read (MBs) and Write (MBs) for NVMe, Solid State, and Hard Disk Drives





Hybrids

- There are also some drives on the market that are a **hybrid of both HDDs and SSDs**
 - This is a cost-cutting measure, which is popular in some laptop lines
- They have the advantages of SSD's up-to a point
 - Write can be cached in the SSD part then spooled to the HDD later
 - Popular read data can be held on the SSD
- Not generally good for intensive tasks like big data though





Redundant Array of Independent Disks (RAID)

- Because individual HDD and SSDs are prone to failure, we often use a dedicated solution to group multiple co-located disks together called **RAID**
 - Makes **multiple disks appear as one disk**
- Both hardware or software options
 - **Hardware**: On-chip RAID controllers
 - **Software**: ZFS, XFS or built-in RAID support at OS level





Striping, Mirroring and Parity

- **Data Striping** is the technique of segmenting logically sequential data, such as a file, so that consecutive segments are stored on different physical storage devices.
 - This means that large reads can be spread over multiple drives
- **Disk Mirroring** is the replication of logical disk volumes onto separate physical hard disks in real time to ensure continuous availability.
- **Parity data** is used by RAID arrays to achieve redundancy.
 - If a drive in the array fails, remaining data on the other drives can be combined with the parity data (using the Boolean XOR function) to reconstruct the missing data.



RAID Levels

- There are different variants of RAID, that make different trade-offs between **reliability, availability, performance, and capacity**
 - RAID 0 consists of striping, but no mirroring or parity.
 - RAID 1 consists of data mirroring, without parity or striping.
 - RAID 2 consists of bit-level striping with dedicated Hamming-code parity.
 - RAID 3 consists of byte-level striping with dedicated parity.
 - RAID 4 consists of block-level striping with dedicated parity.
 - RAID 5 consists of block-level striping with distributed parity.
 - RAID 6 consists of block-level striping with double distributed parity.
- Most storage systems will use either RAID 5 or 6, such that one disk failure will not cause data loss



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